

Lab599 Discovery TX-500

Digital Modes & FT8 Operation on macOS Tahoe

A Comprehensive Engineering Guide: Hardware Cabling, Audio Setup, WSJT-X,
Driver Installation, Firmware Bugs & Fixes, and Thermal Management

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Environment: macOS Tahoe 26.6.2 (25G83) • WSJT-X v3.2.0 260818 improved PLUS • TX-500 Firmware v1.30.00

Abstract

This guide provides an exhaustive, step-by-step manual for operating the **Lab599 Discovery TX-500 QRP HF/6m SDR Transceiver** on **macOS Tahoe 26.6.2 (25G83)** using **WSJT-X v3.2.0 260818 improved PLUS** for FT8 and related digital modes. Topics covered include: the author's verified hardware setup utilizing the **AD-508 USB-C Audio Cable** and the **stock blue USB CAT adapter**, **mandatory FTDI D2XX driver installation** under macOS, radio menu configuration (**Menu 34 CAT Protocol** and **Menu 09 Gain>DIG**), cycling to **DIG Mode** by repeatedly pressing the MODE button, choosing **Kenwood TS-480** in WSJT-X, resolving the critical **85 dB Red Bar Audio Clipping / Saturation** condition via a proven 3-step method, full resolution of the known **DIG mode-switching firmware bugs**, precise audio gain calibration via the TX-500 METR MIC meter and macOS Audio MIDI Setup, RFI mitigation techniques, and practical thermal cooling strategies utilizing external fans and sealed ice-packs.

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Revision History & Acknowledgments

Version	Date	Summary of Changes
1.0	September 16, 2026	Initial official release. Complete manual covering official Lab599 cables (AD-508 USB-C, stock blue CAT adapter), mandatory FTDI D2XX driver installation, Kenwood TS-480 WSJT-X rig selection, MODE button cycling to DIG, 3-step 85 dB audio clipping resolution, firmware bug workarounds, audio calibration, and thermal management.

Acknowledgments & Community Contributions

Special and heartfelt thanks are extended to:

- **BladRunner** — author of the authoritative *TX500MacOS* guide shared in the [Lab599 Telegram Community \(@lab599\)](#), whose early insights into macOS device handling laid the foundation for the community's Apple Silicon digital workflows.
- **Uwe Risse, DG2YCB**, Joe Taylor, K1JT, and the WSJT development team for their relentless work on the WSJT-X improved PLUS suite.
- The [groups.io/g/Lab599](#) and [WSJT-X Groups.io](#) communities for documented findings on CAT protocol timing, Kenwood emulation quirks, and audio routing.

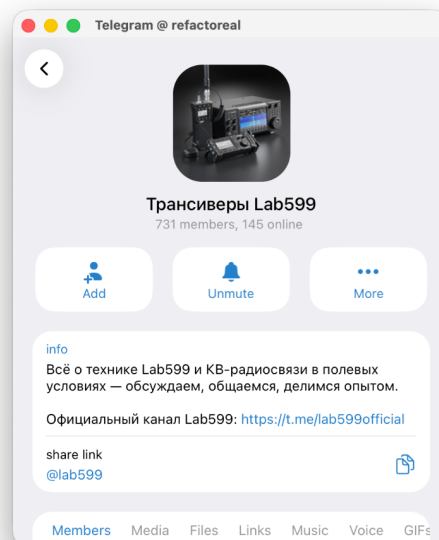


Figure 1: The official Lab599 Telegram Community (@lab599), where BladRunner and other operators regularly collaborate on portable and digital operations.

1 Introduction: The Radio Worth Waiting For

1.1 A Personal Note

“After 7 long weeks of waiting, my TX-500 Discovery radio finally arrived today! Honestly, the delay took so long that I couldn’t even feel as excited as I expected to be when opening it. To top it off, the store canceled 4 items from my order after I placed it because they ran out of stock!

Maybe buying from Russia and shipping to Iran wasn’t the brightest idea after all! But in a world where the shadow of war looms everywhere, there really wasn’t much of a choice.”

— **factoreal (EP2AES)**

Logistics and geopolitical hurdles aside, once unboxed, the Discovery TX-500 reveals itself as a masterclass in portable radio engineering. What follows is the definitive operational blueprint I assembled to spare fellow operators the frustration of undocumented firmware quirks and OS-level communication traps.

1.2 Radio Overview

The **Lab599 Discovery TX-500** is an ultra-compact, all-mode QRP software-defined radio (SDR) engineered specifically for expeditionary, SOTA/POTA, and high-reliability field deployments.

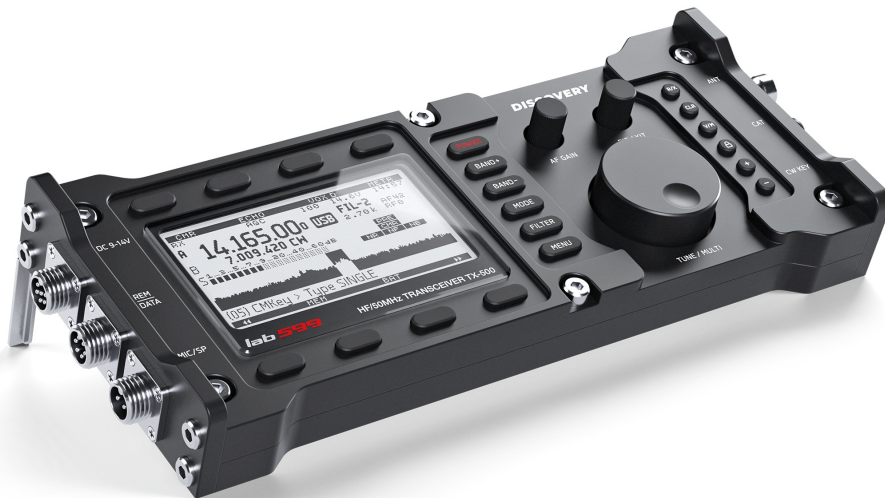


Figure 2: The Lab599 Discovery TX-500 QRP SDR Transceiver: solid CNC-machined aluminum body, integrated 48 kHz panadapter LCD, waterproof GX12 locking connectors, and fold-out kickstands.

The radio’s enclosure is carved from a solid monoblock of aluminum, providing rugged impact resistance and serving as the primary passive heatsink for the transmitter.

The Key Architectural Reality: The TX-500 **does not include an onboard USB soundcard**. Unlike desktop SDR transceivers, all transmit and receive audio travels as analog baseband signals through the locking GX12 REM/DATA port. Consequently, connecting the TX-500 to a

modern Mac requires two distinct physical interfaces: an external audio converter and a serial CAT bridge.

1.3 Technical Specifications

Parameter	Specification
Frequency Range (TX)	160 m through 6 m (all amateur allocations)
Frequency Range (RX)	0.500 MHz – 56.000 MHz (continuous general coverage)
RF Output Power	1.0 W to 10.0 W (continuously adjustable in 0.1 W steps)
Operating Modes	SSB (USB/LSB), CW, DIG, AM, NFM
Supply Voltage	9.0 V to 15.0 V DC external
Dimensions & Weight	228 × 90 × 30 mm, approximately 550 g
Panadapter / Display	48 kHz real-time spectrum/waterfall on 256 × 128 transfective LCD
DSP Architecture	32-bit floating-point DSP engine
Chassis Construction	Fully sealed CNC-machined anodized aluminum
Physical Connectors	GX12 aviation-type threaded locking connectors
CAT Control Emulation	Kenwood command set (Menu 34: TS2000; WSJT-X Rig: Kenwood TS-480)

2 System Architecture: Two Independent Channels

Operating FT8 on macOS requires **two independent physical and logical channels**:

- 1. Analog Audio Channel (REM/DATA Port, GX12 7-Pin):** Carries bidirectional analog audio. Transmit audio (synthesized FT8 tones) travels from macOS to the TX-500 modulator. Receive audio from the receiver detector travels from the radio to macOS for DSP decoding in WSJT-X.
- 2. Serial CAT Control Channel (CAT Port, GX12 4-Pin):** Carries bidirectional serial data. WSJT-X continuously polls the transceiver for dial frequency and mode, commands VFO frequency shifts for “Fake It” split operation, and asserts transmit state via CAT PTT commands.

Both Connections Are Strictly Required

You cannot operate FT8 without both links active simultaneously. Audio without CAT prevents automated frequency tracking and PTT triggering. CAT without audio leaves WSJT-X with no RF signals to decode.

3 Hardware Cables & Interface Options

3.1 Author’s Verified Real-World Hardware Setup

For the author’s primary tested station (EP2AES) running macOS Tahoe on an Apple Silicon MacBook, the setup uses the following optimal cabling:

- **Digital Audio Link (REM/DATA Port):** Official **AD-508 USB-C Audio-Data Cable** connected directly to the MacBook’s native **USB-C** port. This routes full-duplex digital audio directly through the GX12 7-pin port without requiring any external soundcard box, audio dongle, or USB hub.
- **CAT Serial Control Link (CAT Port):** The **original blue-colored USB adapter included inside the TX-500 radio box** (standard USB-A CAT serial bridge with GX12 4-pin connector), connected to the Mac (directly or via a passive USB-A to USB-C adapter).

This verified configuration is the most compact, reliable, and electrically stable arrangement for macOS Tahoe.

3.2 CAT Control: AD-502 / Stock Blue Adapter / AD-514

The CAT adapter (whether the stock blue adapter from the TX-500 box or the black AD-502) terminates on one end in a 4-pin GX12 connector and on the other in a standard USB plug containing an FTDI USB-to-UART serial bridge.



Figure 3: *CAT-USB Adapter: 4-pin GX12 to USB serial bridge for CAT commands and PTT control.*

The **AD-514** is the direct USB-C variant of the CAT adapter.

3.3 Audio Interface Options (REM/DATA Port)

Option A: AD-508 USB-C Audio-Data Cable (Author's Choice — Direct USB-C)

The **AD-508** connects the 7-pin REM/DATA port directly to a native USB-C port with built-in DAC/ADC hardware. It is the cleanest and most direct setup for modern MacBooks.

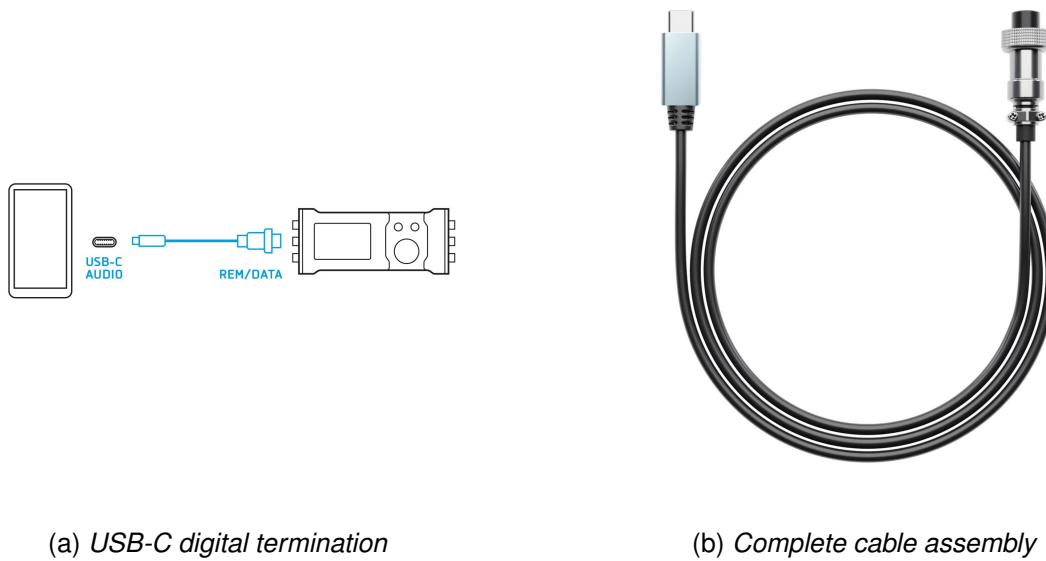


Figure 4: AD-508 USB-C Audio-Data Cable: direct connection for Apple Silicon MacBooks.

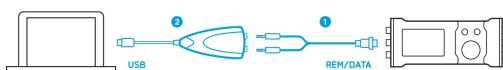
Option B: AD-509 USB PnP Audio Interface (Standard USB-A)

The **AD-509** features an inline C-Media USB audio codec integrated into the 7-pin GX12 cable, terminating in a USB-A connector.

Комплектация



Схема подключения



(a) Inline USB soundcard module

(b) Complete kit with GX12 connector

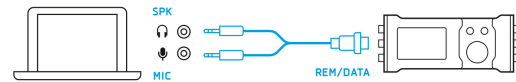
Figure 5: AD-509 USB PnP Audio Interface: plug-and-play USB-A audio solution.

Option C: AD-506 PC Audio-Data Cable (Analog 3.5 mm Breakout)

The **AD-506** provides two 3.5 mm TRS analog audio jacks (Speaker Out and Mic In) for use with third-party external USB soundcards.



(a) GX12 to dual 3.5 mm breakout



(b) Speaker and mic connectors

Figure 6: AD-506 PC Audio Cable: requires an external USB soundcard on macOS.

4 Mandatory FTDI D2XX Driver Installation

4.1 Why This Step Is Mandatory

On macOS (especially Apple Silicon M-series on Sonoma, Sequoia, and Tahoe), the default Apple USB serial driver often fails to instantiate the correct callout device node (`/dev/cu.usbserial-*`) for the FTDI chip inside the CAT adapter. **Installing the official FTDI D2XX runtime is strictly required** to ensure robust, non-blocking serial communication in WSJT-X.

Do Not Skip Driver Installation

If you skip this step, WSJT-X will almost certainly produce a “Rig Failure” or red “Test CAT” error, even if the cable is plugged in firmly.

4.2 Step-by-Step Installation Procedure

Step 1. Download the Driver: Obtain the latest macOS D2XX driver disk image (D2XX1.4.35.dmg or newer) from the official FTDI website: ftdichip.com/drivers/d2xx-drivers/.

Step 2. Extract Release Files: Double-click the downloaded .dmg file to mount it. Drag the folder named `release` directly onto your macOS **Desktop**.

Step 3. Open Terminal: Launch `Terminal.app` from `/Applications/Utilities/`.

Step 4. Execute Installation Commands: Run the following terminal commands in exact sequence:

```
# 1. Ensure system destination directories exist
sudo mkdir -p /usr/local/lib
sudo mkdir -p /usr/local/include
```

```
# 2. Copy the dynamic shared library (verify exact filename in release/build
/)
sudo cp ~/Desktop/release/build/libftd2xx.1.4.35.dylib /usr/local/lib/
libftd2xx.1.4.35.dylib

# 3. Create the canonical symbolic link
sudo ln -sf /usr/local/lib/libftd2xx.1.4.35.dylib /usr/local/lib/libftd2xx.
dylib

# 4. Copy required C/C++ header files
sudo cp ~/Desktop/release/ftd2xx.h /usr/local/include/ftd2xx.h
sudo cp ~/Desktop/release/WinTypes.h /usr/local/include/WinTypes.h

# 5. Verify the installation
ls -la /usr/local/lib/libftd2xx*
```

Listing 1: Installing FTDI D2XX Runtime on macOS

Driver Verification

If Step 5 lists libftd2xx.dylib pointing to the versioned dylib, the driver is correctly installed. Disconnect and re-insert the CAT adapter into your Mac.

5 Radio Menu Configuration (Mandatory)

Configure these settings on the TX-500 hardware before starting WSJT-X.

5.1 Menu Setup Steps

Step 1. Power on the TX-500.

Step 2. Press the **MENU** push-button on the front panel.

Step 3. Rotate the **TUNE/MULTI knob** to **Menu 34 (CAT Protocol)**:

- Press the knob to activate editing.
- Rotate until the display reads TS2000.
- Press **MENU** to confirm.

Step 4. Rotate to **Menu 35 (CAT Rate)**:

- Ensure it is set to 9600 (factory default).

Step 5. Rotate to **Menu 09 (Gain>DIG)**:

- Set to an initial baseline of 30 (range 20–40).

Step 6. Press **MENU** to exit and save all menu parameters.

Step 7. Switch to DIG Mode: Press the front-panel **MODE** button **repeatedly** until the operating mode cycles through (LSB → USB → CW → DIG...) and the display clearly shows DIG.

Always Operate in DIG Mode — Repeatedly Press the MODE Button

To select DIG mode on the TX-500, you must press the front-panel **MODE** button several times until the DIG label appears on the LCD. In DIG mode, the radio isolates the microphone line and enables audio routing exclusively through the REM/DATA port. Operating in USB or LSB mode bypasses the digital interface, resulting in zero transmit RF and decoding failure.

6 macOS Device Verification

6.1 Audio Interface Verification

With the AD-508 or AD-509 connected:

1. Open **System Settings** → **Sound**.
2. Verify **USB Audio Device** is listed under both **Output** and **Input**.

6.2 macOS Privacy: Microphone Permission

Critical: Grant WSJT-X Microphone Access

On macOS Sonoma, Sequoia, and Tahoe, third-party apps cannot access USB audio inputs without explicit user authorization. Without this permission, the WSJT-X waterfall remains completely blank.

1. Open **System Settings** → **Privacy & Security** → **Microphone**.
2. Ensure the toggle next to **WSJT-X** is **ON**.
3. If WSJT-X is not listed, launch it once to trigger the OS prompt, grant permission, and restart WSJT-X.

6.3 Serial Port Device Identification

Open Terminal and execute:

```
ls -al /dev/cu.usbserial*
```

Listing 2: Identifying the CAT Serial Device Node

Typical response:

```
crw-rw-rw-  1 root  wheel  21,4   Sep 16 14:30 /dev/cu.usbserial-11140
```

Record this path (e.g., /dev/cu.usbserial-11140) for entry into WSJT-X.

6.4 Audio MIDI Setup Configuration: Calibrated Volume Levels

Launch **Audio MIDI Setup** (/Applications/Utilities/Audio MIDI Setup.app).

Extensive testing and isolation of all hardware and software variables on macOS Tahoe revealed two vital calibration rules that must be strictly maintained for the USB audio device (e.g., AD-508 / AD-509):

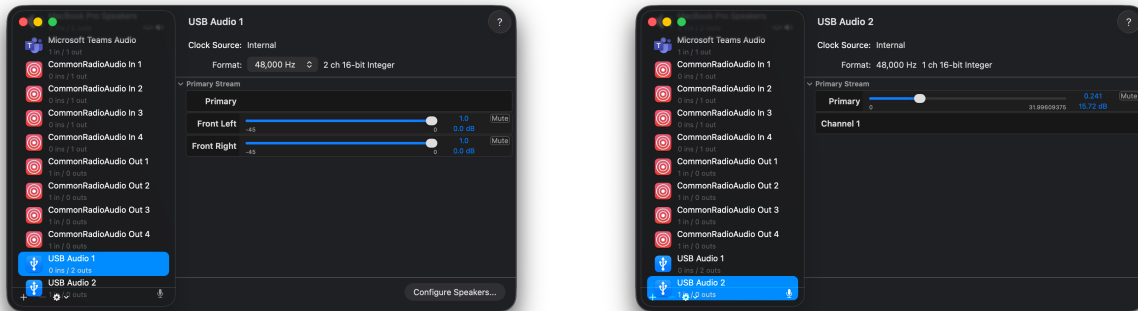
1. Output Volume MUST Be Set to 100% (1.0 / 0.0 dB):

After extensive troubleshooting regarding why the radio keyed into transmit but showed **0 Watts RF output power**, testing every single element in the transmission chain conclusively proved that the macOS Output Volume for the USB soundcard must be set to **100%** (slider value 1.0, 0.0 dB, as shown in Figure 7a). Anything less fails to provide sufficient analog drive voltage to the TX-500 modulator via the REM/DATA port, leaving the radio in transmit with zero RF power.

2. Input Volume MUST Be Set to at Most 24% (0.241 / 15.72 dB):

To prevent soundcard ADC saturation and audio clipping (which causes the WSJT-X left dB bar to stick at an overloaded **85 dB red**, preventing any signals from decoding), the macOS

Input Volume must be set to **no more than 24%** (slider value ≤ 0.241 , as shown in Figure 7b). This reliably stabilizes the receive noise floor inside the optimal 30–50 dB green decoding range.



(a) Output volume at 100% (1.0 / 0.0 dB): critical for RF power

(b) Input volume at $\leq 24\%$ (0.241): critical to prevent clipping

Figure 7: Audio MIDI Setup calibration on macOS Tahoe: 100% Output ensures full RF output drive, and $\leq 24\%$ Input keeps WSJT-X in the optimal green decoding zone.

7 WSJT-X Software Configuration

This setup is verified against **WSJT-X v3.2.0 260818 improved PLUS**.

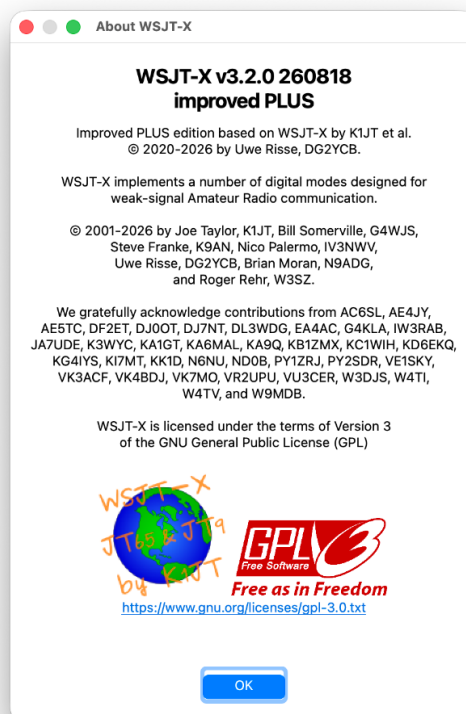


Figure 8: Verified software environment: WSJT-X v3.2.0 260818 improved PLUS edition.

7.1 General Tab

In **WSJT-X** → **Settings** → **General**:

- **My Call:** Enter your amateur callsign (e.g., EP2AES).
- **My Grid:** Enter your Maidenhead locator (e.g., LM55).
- **Uncheck:** “Allow Tx frequency changes while transmitting”.

7.2 Radio Tab (Most Critical Settings)

Navigate to **Settings** → **Radio** and configure each parameter exactly as specified:

Field	Value and Technical Rationale
Rig	Kenwood TS-480 — MANDATORY selection in WSJT-X . While Menu 34 on the radio is set to TS2000, selecting Kenwood TS-480 in WSJT-X provides the most reliable polling stability, frequency tracking, and CAT PTT response without erratic errors.
Serial Port	/dev/cu.usbserial-XXXX (as determined via Terminal).
Baud Rate	9600 (matches Menu 35).
Data Bits / Stop Bits	8 Bits / 2 Stop Bits (try 1 if communication errors occur).
Handshake	None .
Transmit Audio Source	Rear/Data (routes transmit audio to REM/DATA).
Mode	None — MANDATORY: Prevents Firmware Bug #1 .
Split Operation	Fake It (shifts VFO on TX to ensure optimal audio passband).
PTT Method	CAT (reliable serial PTT assertion).

Firmware Bug #1: DIG → USB Mode Switching on Transmit

The Symptom: When WSJT-X starts transmitting, the TX-500 LCD jumps from DIG to USB. The radio enters transmit, but produces **0 Watts RF output**.

The Root Cause: When WSJT-X Mode is set to Data/Pkt or USB, it transmits an automated CAT mode command at every transmit cycle. The TX-500 firmware (through v1.30.00) misinterprets this Kenwood command and forces the radio into SSB voice mode, disconnecting the REM/DATA port.

The Proven Fix: Set **Mode: None** in WSJT-X Settings. Manually set the radio to DIG mode on the front panel by repeatedly pressing the **MODE** button before each session. WSJT-X will omit mode commands, and the radio will remain locked in DIG mode throughout all transmissions.

Verifying CAT Connection

Click **Test CAT**. The button must turn **Green**. Then click **Test PTT**; the radio should briefly switch to transmit and return to receive.

7.3 Audio Tab

In **Settings** → **Audio**:

- **Soundcard Input:** Select USB Audio Device.
- **Soundcard Output:** Select USB Audio Device.

8 Audio Gain Calibration

8.1 Receive Audio Calibration & Resolving the 85 dB Red Bar Saturation Bug

Upon first connecting the audio cable to macOS, a very common and critical issue frequently occurs:

Audio Clipping & Sound Card Saturation (85 dB Red Bar)

The Symptom: The vertical dB meter on the left side of the main WSJT-X window reads **85 dB** and turns **solid red**.

The Technical Cause: An 85 dB reading with a red indicator means the incoming audio signal is excessively strong (*Audio Clipping*), completely saturating the soundcard's ADC input. In this saturated condition, the WSJT-X DSP decoder cannot distinguish weak FT8 tones, **completely preventing any signals from decoding**.

Target Ideal Level: In receive mode (whether the channel is quiet or normal signals are present), the input level should hover comfortably between **30 dB and 50 dB**, and the vertical indicator bar must remain **solid green**.

The 3-Step Resolution Procedure (3 Minutes Total)

Follow these three consecutive 1-minute steps to bring the audio level into the optimal green zone:

Step 1. Reduce Input Volume in macOS (1 Minute):

- In macOS, navigate to **System Settings** → **Sound** (or launch **Audio MIDI Setup**).
- Select the **Input** tab and your DATA soundcard (USB Audio 2 / USB Audio Device).
- Drag the **Input volume** slider down until it is at **at most 24%** (slider value ≤ 0.241 , see Figure 7b). Setting the slider above 24% pushes the soundcard ADC into immediate saturation.
- *Verification:* Watch the vertical meter on the left of WSJT-X. The 85 dB number should immediately start dropping, and the color will begin changing away from red toward green.

Step 2. Adjust Audio Output Level on the TX-500 (1 Minute):

- If lowering the macOS input volume does not lower the noise floor sufficiently, reduce the audio output level directly from the TX-500.
- On the radio, adjust **Menu 09 (Gain>DIG)** or use the front-panel **AF / Line Out** adjustment to decrease the audio output level sent to the REM/DATA port.
- *Verification:* With the reduction of the radio volume, the dB reading in WSJT-X responds immediately and drops further.

Step 3. Stabilize Audio Level Within the Green Zone (30 dB to 50 dB) (1 Minute):

- Fine-tune the macOS input volume and the radio output so that during the reception of normal signals, the meter reading remains steadily between **30 dB and 50 dB**, and the indicator bar is **solid green**.
- *Verification:* Once the bar is green, at the conclusion of each 15-second FT8 cycle, decoded messages will immediately appear in the **Band Activity** window!

8.2 Transmit Audio Level Calibration & Resolving the 0 W RF Output Bug

Vital: macOS Output Volume Must Be 100% to Prevent 0 W RF Output

After extensive troubleshooting regarding why the radio entered transmit (red LED illuminated, PTT engaged via CAT) but showed **0 Watts RF power output** on the inline wattmeter, testing every single element and cable in the chain conclusively proved that the macOS Output Volume for the USB soundcard **must be set to 100% (1.0 / 0.0 dB)** in Audio MIDI Setup (Figure 7a). At lower volume settings, the audio drive voltage sent to the TX-500 REM/DATA port is insufficient to excite the PA stages, resulting in zero transmitted RF power!

- Step 1.** Connect a matched 50 Ω dummy load or resonant antenna.
- Step 2.** Ensure macOS Output Volume is set to **100% (1.0 / 0.0 dB)** in Audio MIDI Setup.
- Step 3.** On the TX-500, set the display meter to **MIC** (via front panel METR button or Menu 23).
- Step 4.** In WSJT-X, set the right-side **Pwr** slider to approximately **−3 dB**.
- Step 5.** Click the **Tune** button in WSJT-X to transmit a steady tone.
- Step 6.** Observe the **MIC** bar graph on the TX-500 screen:
 - Adjust **Menu 09** until the meter deflects to **approximately 3/4 scale**.
 - **Do not saturate or pin the meter to maximum scale.**
- Step 7.** Verify RF power on the inline wattmeter (e.g., NISSEI DG-503 MAX).
- Step 8.** Click **Tune** again to return to receive.

Avoid Audio Overdrive

Overdriving the digital audio input causes severe ALC compression and intermodulation splatter, broadening your signal and causing interference across adjacent FT8 slots.

9 Comprehensive Bug Reference & Troubleshooting

Issue	Root Cause	Verified Solution
WSJT-X shows 85 dB and red bar (Audio Clipping)	Incoming RX audio excessively high; soundcard ADC saturated.	Follow 3-step fix: (1) Set macOS Input Volume to ≤24% in Audio MIDI Setup; (2) Lower TX-500 Menu 09; (3) Target 30–50 dB solid green bar.
Radio keys into TX but produces 0 Watts RF power	(a) Insufficient macOS audio drive, or (b) DIG jumped to USB voice mode.	(1) Set macOS Output Volume to 100% (1.0) in Audio MIDI Setup; (2) Set WSJT-X Mode to None ; (3) Repeatedly press MODE to lock in DIG .
Decoding fails after Mac wakes from sleep	macOS resets USB audio sample rate from 48000 Hz to 44100 Hz.	Reset sample rate in Audio MIDI Setup to 48000 Hz. Disable Mac system sleep during operation.
Waterfall stops decoding after 20 minutes	CoreAudio buffer overflow during continuous USB TX/RX cycles.	Connect audio cable directly to Mac (bypass unpowered hubs). Restart WSJT-X.
RX audio drops out immediately after TX	macOS Privacy subsystem momentarily blocks microphone input on state change.	Ensure Microphone Permission is granted. Launch WSJT-X from /Applications/, not Terminal.
CAT port locked after unexpected app crash Rapid TX/RX relay chattering	macOS locks serial device descriptor in kernel. Common-mode RF circulating on USB and CAT cables corrupts UART packets.	In Terminal run <code>sudo killall -9 wsjtx</code> . Unplug and re-insert CAT adapter. Add 1:1 RF choke to feedline; install Type-43 snap-on ferrites on USB cables; run Mac on battery.

10 RF Power Monitoring & Inline Wattmeter

10.1 Duty Cycle Considerations

FT8 transmits at a continuous 100% key-down duty cycle for 12.6-second intervals. With standard 50% TX/RX sequences:

- **Without supplemental cooling:** Limit continuous power to **3.0 W to 5.0 W**.
- **With external cooling (fan or ice-pack):** Safe sustained operation at **8.0 W to 10.0 W**.

10.2 Monitoring with NISSEI DG-503 MAX

Insert a digital inline wattmeter (such as the **NISSEI DG-503 MAX**) between the radio BNC output and the antenna feedline:

- Continuously observe forward power, reflected power, and real-time SWR.
- Maintain antenna SWR at **1.5:1 or better**.
- Never operate at 10 W full power if SWR exceeds 2.0:1.

11 Thermal Management Strategies

11.1 Thermal Physics of the TX-500

Because the TX-500 is completely sealed against moisture, it relies entirely on conduction from the PA transistors to the aluminum chassis, followed by natural convection to surrounding air. During prolonged 10 W FT8 sessions, chassis temperatures can exceed 55 °C, triggering internal thermal foldback.

11.2 Method 1: External Cooling Fan (Recommended for Shack Operation)

Directing forced airflow across the chassis is the most practical and effective cooling method:

- Use a small **12 V or 5 V USB desk fan** placed 10–20 cm from the radio.
- Aim the airflow directly along the bottom cooling ribs and top heat-dissipation surfaces.
- **Observed result:** Chassis temperature drops by **15 °C to 20 °C**, allowing indefinite 10 W FT8 operation without thermal throttling.

11.3 Method 2: Sealed Ice-Pack Cooling (Field Deployments)

For portable or field operation where electric fans are impractical:

- Place an ice pack or crushed ice inside a **heavy-duty, double-sealed freezer bag**.
- Rest the bag directly against the flat aluminum underside of the radio chassis.
- The high thermal conductivity of the aluminum transfers heat rapidly into the ice mass.

Condensation Precaution

The TX-500 internal cavity is fully sealed. Condensation occurs only on the external aluminum surface. Always place a dry cloth beneath the setup to absorb condensation, and wipe the exterior connectors periodically.

12 RFI Mitigation Best Practices

High common-mode RF currents are the leading cause of CAT disconnects and USB audio drops on portable setups:

1. **Common-Mode Choke at Antenna Feedpoint:** Wind 8 to 10 turns of RG-316 or RG-58 coax through an **FT-240-43 ferrite toroid** immediately adjacent to the antenna or radio output.
2. **Ferrite Chokes on Data Cables:** Place snap-on Fair-Rite Type-43 ferrites on both ends of the CAT cable and audio cable.
3. **Run MacBook on Battery:** Disconnect the Mac mains charger during digital sessions to eliminate AC ground loops.

13 Pre-Session Operational Checklist

Review this checklist before starting any digital operation:

- ☐ Connect **AD-508** (audio) to REM/DATA port and Mac USB-C.
- ☐ Connect **stock blue CAT adapter** to CAT port and Mac USB.
- ☐ Connect resonant antenna or 50 Ω dummy load to BNC ANT jack.
- ☐ Power on the TX-500 (12.0–13.8 V DC).
- ☐ Verify **Menu 34 = TS2000** and **Menu 35 = 9600**.
- ☐ Press front panel **MODE** button **repeatedly** until screen indicates DIG.
- ☐ Confirm Audio MIDI Setup: Output = **100% (1.0 / 0.0 dB)** and Input = **at most 24% (0.241)** at **48000.0 Hz 16-bit**.
- ☐ Verify WSJT-X has **Microphone Permission** under System Settings.

- ☐ In WSJT-X Radio Settings: Rig = Kenwood TS-480, Baud = 9600, Mode = None, Split = Fake It, PTT = CAT.
- ☐ Click **Test CAT** → verify indicator turns **Green**.
- ☐ **Verify RX Level:** Ensure WSJT-X left dB bar reads **30 dB to 50 dB (solid green)**. If reading **85 dB red**, reduce macOS input volume to $\leq 24\%$ in Audio MIDI Setup and lower Menu 09.
- ☐ Click **Tune** in WSJT-X; verify METR MIC reads $\approx 3/4$ scale on TX-500 LCD and observe RF power on inline wattmeter.
- ☐ Position external fan or ice-pack if operating above 5 W continuous output.
- ☐ Enable **Auto Seq** and **Enable Tx** in WSJT-X. Begin operating!

14 Conclusion

When properly configured with the AD-508 USB-C audio cable, verified FTDI drivers, Kenwood TS-480 rig selection, and calibrated audio levels, the Lab599 Discovery TX-500 represents one of the finest portable FT8 stations available for macOS Tahoe.

Feedback & Community Contact

For technical feedback, corrections, or field notes, contact the author:
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73 and Good DX from EP2AES!